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1.

a) False.

Bootstrap distribution از resampling کردن از اول نمونه‌ی اولیه
که از جامعه برداشتیم ایجاد می‌شود، نه خود جامعه.

b) True.

توضیح خود همین مدل تستی، وقتی فرض ما رد بشه، یعنی حداقل یکی از
جفت گروه‌ها میانگین یکسانی ندارند ولی نمی‌دونیم کدوم جفت گروه.

c) False.

سایز resample ها معمولاً به اندازه‌ی همون sample اولیه است که
اول برداشتیم، سایزش کوچکتر ترجیح نمی‌شود. نمونه‌ی اصلی

d) True.

در paired analysis اول ما تفاوت بین گروه‌ها رو داریم (بین هر جفت گروه)
بعد صرفاً از اون difference استفاده می‌شه برای inference.

e) False.

بین واریانس درون یک گروه ANOVA و میزان نویز (واریانس)
یک رابطه‌ی مستقیم وجود دارد، هرچقدر واریانس درون گروه‌ها بیشتر باشه، نشان دهنده‌ی
وجود نویز بیشتری درون گروه‌ها است.

f) True.

بله این یک T-test paired هستش باشه چون $n \geq 30$ هست میشه از Z هم استفاده کرد.

g) True.

در paired anylasis، تمرکز ما بر اساس تفاوت بین دو مشاهده برای هر جفت از داده های ما می شود، پس عبارت درستی است.

h) False.

خیر، برای جلوگیری از خطای نوع دوم، باید مقدار β را کاهش داد نه افزایش. مقدار β احتمال خطای نوع دوم را نشان میدهد، کاهش β منجر می شود به افزایش power و $1-\beta$ که نشان دهنده‌ی احتمال صحیح رد کردن فرض نادرست می باشد.

2.

a)

$$\bar{x} = \frac{76+84+69+92+58+89+73+97+85+77}{10} = \frac{800}{10} = 80$$

$$S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

$$= \frac{(76-80)^2}{9} + \frac{(84-80)^2}{9} + \frac{(69-80)^2}{9} + \frac{(92-80)^2}{9} +$$

$$\frac{(58-80)^2}{9} + \frac{(89-80)^2}{9} + \frac{(73-80)^2}{9} + \frac{(97-80)^2}{9} + \frac{(85-80)^2}{9} +$$

$$\frac{(77-80)^2}{9} = \frac{1234}{9} = 137.11 \Rightarrow S = 11.71$$

$$SE = \frac{S}{\sqrt{n}} = \frac{11.71}{\sqrt{10}} = 3.7$$

b) Margin of error $= t^* \times SE$

از دیدگاه شروط برای استفاده، برقرار است و شرط مستقل بودن،

(random sampling/assignment و n از 10% کل جامعه کوچکتر است)

فقط شرط $n \geq 30$ بودن برقرار نیست اینجا ما از t-test استفاده می‌کنیم،

$df = n - 1 = 9$

if we consider $\alpha = 0.1$ we have $t^* = 1.83$

R: qt (0.05 , df = 9)

– 1.833113

$ME = 1.83 \times 3.7 = 6.77$

c) $\bar{x} - ME < \mu < \bar{x} + ME$

$80 - 6.77 < \mu < 80 + 6.77 \Rightarrow 73.23 < \mu < 86.77$

$(73.23 , 86.77)$

d) with 90% confidence, we can say that the mean score of all 500 students in this statistics class lies between 73.23 and 86.77.

3.

a) $H_0: \mu = 8$ hours (NewYorkers sleep 8 hours a night on avg)

$H_A: \mu < 8$ hours (NewYorkers sleep less than 8 hours a night on average)

b)

Independence:

- Sampled observations are 25 random independent Newyorkers. ☑
- The Sampling is without replacement, and 25 is less than 10 % of the population of New York. ☑

Sample size / Skew:

- The amount of sleep that people get per day is a natural trait that is normally distributed.
For this size sample, slight Skew is acceptable, and the min/max suggests there is not much Skew in the data
- The size of the sample is small and not larger than 30 ($n < 30$)

Therefore, we use student's t– Distribution:

$$SE = \frac{S}{\sqrt{n}} = \frac{0.77}{5} = 0.154$$

$$T = \frac{\text{observation} - \text{Null}}{SE} = \frac{7.73 - 8}{0.154} = -1.75$$

$$df = n - 1 = 25 - 1 = 24$$

c) p-value = 0.0464

R:

```
pt(-1.75, df = 24)
0.04644754
```

if we consider $\alpha = 0.05$ we can conclude that p-value < α. Therefore, the data provide enough evidence to reject the H_0 meaning that the data suggests that New Yorkers sleep less than 8 hours a night on average.

d) we know that if the p-value is less than the significance level, we reject the null hypothesis and because p-value = 0,0464 and is less than $\alpha = 0,05$, we reject the H_0.

e) No. using p-value we concluded that H_0 has been rejected meaning that 8 will not be present in our CI.

For 90% confidence interval we have $t^* = 1.71$

R:

```
qt(0.95, df = 24)
1.710882
```

$$\bar{x} - t^* SE < \mu < \bar{x} + t^* SE$$

$$7.73 - 1.71 \times 0.154 < \mu < 7.73 + 1.71 \times 0.154 \Rightarrow$$

$$7.4605 < \mu < 7.9995 \qquad (7.4605, 7.9995)$$

Therefore, the H_0 hypothesis will be rejected, and 8 is not in 90% confidence interval.

4. we need to calculate the difference between mean of two groups.

$\mu_{intensive} = \mu_{paced} \Rightarrow \mu_{intensive} - \mu_{paced} = 0$

$H_0: \mu_{intensive} - \mu_{paced} = 0$ (There is no difference)

$H_A: \mu_{intensive} - \mu_{paced} \neq 0$ (There is difference)

$$SE_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} = \sqrt{\frac{41.4736}{12} + \frac{56.5504}{10}} = 3.02$$

$df = \min(n_1 - 1, n_2 - 1) = 9$

we can calculate the exact value:

$$C = \frac{\frac{S_1^2}{n_1}}{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} = \frac{\frac{41.4736}{12}}{\frac{41.4736}{12} + \frac{56.5504}{10}} = 0.38$$

$$df = \frac{(n_1 - 1) \times (n_2 - 2)}{(n_2 - 1) \times C^2 + (1 - C^2)(n_1 - 1)} =$$

$$\frac{9 \times 11}{11 \times (0.38)^2 + (1 - 0.38)^2 \times 9} = 19.604397225 8 \simeq \underline{\underline{20}}$$

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{3.02} = \frac{46.31 - 42.79}{3.02} = 1.1655$$

2 * pt(1.165, df = 20, lower.tail = FALSE)
0.2577267

: R

p-value = 0.25

if we consider $\alpha = 0.05$ we calculate that p-value $> \alpha$
and we fail to reject H_0.

5.

effect size $= \delta = \bar{x} - \mu = 0.5$, $\beta = 0.2$,

$\alpha = 0.5$ $z = ۱,۹۶ + ۰,۸۴ = ۲,۸$

R

qnorm (0.975)	qnorm (0.8)
1.959966	0.8416212
≈ 1.96	≈ 0.84

, $sd = 2.2$, $n = ?$

$$z = \frac{\text{observation} - \text{null}}{SE} \Rightarrow SE = \frac{\delta}{z}$$

$$SE = \sqrt{\frac{S_1^۲}{n} + \frac{S_۲^۲}{n}} \Rightarrow \frac{\delta^۲}{z^۲} = \frac{۲S^۲}{n} \rightarrow n = ۲ \times S^۲ \times z^۲ / \delta^۲$$

$$۲ \times (۲,۲)^۲ \times (۲,۸)^۲ / ۰,۵^۲ = 303.56 \rightarrow \boxed{n = 304}$$

6.

First we need to calculate the difference between each group.

Before	After	Difference
25	27	25 − 27 = −2
25	29	−4
27	37	−10
44	56	−12
30	46	−16
67	82	−15
53	57	−4
52	61	−9
53	80	−27
60	59	1
28	43	−15

$H_0 : \mu_{Diff} = 0$ (There is no difference)

$H_A : \mu_{Diff} \neq 0$ (There is difference)

$$\bar{x} = \frac{-2-4-10-12-16-15-4-9-27+1-15}{11} = -10.27$$

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

$$= \frac{(-2+10.27)^2}{10} + \frac{(-4+10.27)^2}{10} + \frac{(-10+10.27)^2}{10} +$$

$$\frac{(-12+10.27)^2}{10} + \frac{(-16+10.27)^2}{10} + \frac{(-15-10+27)^2}{10} +$$

$$\frac{(-4+10.27)^2}{10} + \frac{(-9+10.27)^2}{10} + \frac{(-27+10.27)^2}{10} +$$

$$\frac{(1+10.27)^2}{10} + \frac{(-15+10.27)^2}{10} = \frac{636.1879}{10} = 63.62$$

$$s = \sqrt{63.62} = 7.97$$

$$df = n-1 = 11-1 = 10$$

$$SE = \frac{s}{\sqrt{n}} = \frac{7.97}{\sqrt{11}} = 2.40$$

$$T = \frac{\text{observed} - \text{null}}{SE} = \frac{-10.27-0}{2.40} = 4.28$$

R:

P-value = 0.0008056962

$\boxed{\text{pt(4.28, df = 10, lower.tail = FALSE)} \quad 0.0008056962}$

Therefore, p-value < α and we reject the null hypothesis.

7. Check conditions:

1. independence → within group → random sample / assignment ☑

→ $n < 10\%$ of population ☑

→ between group → گروه ها باید ازهم مستقل باشند ☑

2. approximately normally توزیع نزدیک به نرمال ☑

3. Equal variance variability ☑

a)

H_0: The mean height is the same across all the mentioned countries.

$$\mu_{USA} = \mu_{UK} = \mu_{India}$$

H_A: The mean height differs between at least one pair of countries.

b)

		DF	Sum SQ	Mean SQ	F-value	pr (>F)
Group	Class	2	1.75	0.857	0.02683461117	0.9735
Error	Residuals	21	684.75	32.607		
	Total	23	686.5			

$$\bar{y}_{usa} = \frac{180 + 183 + 172 + 178 + 169 + 179 + 178 + 180}{8} = 177.375$$

$$\bar{y}_{uk} = \frac{185 + 181 + 180 + 179 + 164 + 173 + 180 + 178}{8} = 177.5$$

$$\bar{y}_{India} = \frac{170 + 183 + 180 + 175 + 181 + 183 + 176 + 167}{8} = 176.875$$

$$\bar{y} = \frac{176.875 + 177.5 + 177.375}{3} = 177.25$$

$$SSG = \sum_{j=1}^{k} n_j \times (\bar{y}_j - \bar{y})^2 =$$

$$8\,(177.375 - 177.25)^2 + 8\,(177.5 - 177.25)^2 +$$
$$8\,(176.875 - 177.25)^2 = 1.75$$

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2 = (180 - 177.25)^2 + (183 - 177.25)^2$$
$$+ (172 - 177.25)^2 + (169 - 177.25)^2 + (179 - 177.25)^2 +$$
$$(178 - 177.25)^2 + (169 - 177.25)^2 + (179 - 177.25)^2 +$$
$$(178 - 177.25)^2 + (180 - 177.25)^2 + (185 - 177.25)^2 +$$
$$(187 - 177.25)^2 + (180 - 177.25)^2 + (179 - 177.25)^2 +$$
$$(164 - 177.25)^2 + (173 - 177.25)^2 + (180 - 177.25)^2 +$$
$$(178 - 177.25)^2 + (170 - 177.25)^2 + (183 - 177.25)^2 +$$
$$(180 - 177.25)^2 + (175 - 177.25)^2 + (187 - 177.25)^2 +$$
$$(173 - 177.25)^2 + (176 - 177.25)^2 + (167 - 177.25)^2 =$$
$$= 686.5$$

$$SSE = SST - SSG = 686.5 - 1.75 = 684.75$$

Degrees of Freedom :

Total : $df_T = n - 1 = 24 - 1 = 23$

Group : $df_G = K - 1 = 3 - 1 = 2$

Error : $df_E = df_T - df_G = 23 - 2 = 21$

Mean squares:

Group: $MSG = \frac{SSG}{df_G} = \frac{1.75}{2} = 0.875$

Error: $MSE = \frac{SSE}{df_E} = \frac{684.75}{21} = 32.607$

F - statics:

$$F = \frac{MSG}{MSE} = \frac{0.875}{32.607} = 0.02683461117$$

Calculating p-value:

p-value = 0.9735

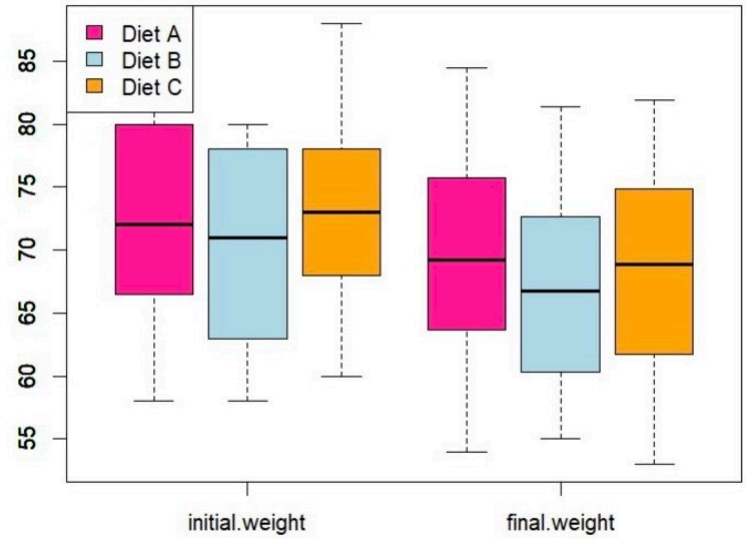
R:

pf(0.02683461117, 2, 69, lower.tail = FALSE)
0.9735324

C) p-value is large (p-value $> \alpha$), we fail to reject H_0.

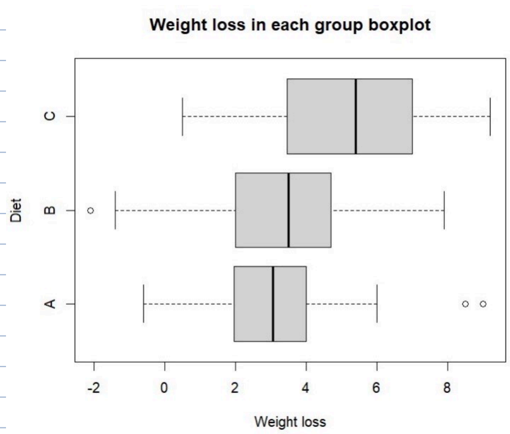
8.

a) The weights boxplot before and after the diet are like the below figure:



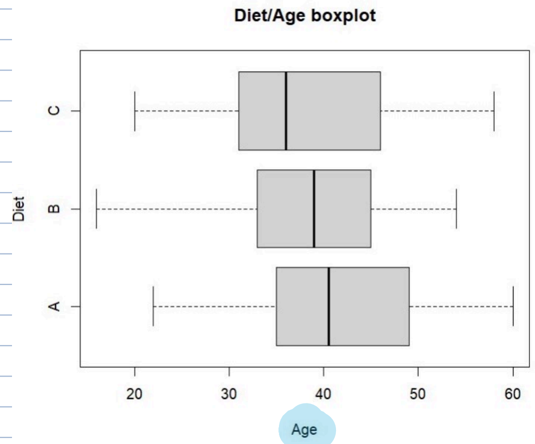

```
> boxplot(df[,c(6,7)], boxfill = NA, border = NA) #invisible boxes - only axes and plot area
> boxplot(df[df$diet.type=="A", c(6,7)], xaxt = "n", add = TRUE, boxfill="deeppink",
+         boxwex=0.25, at = 1:ncol(df[,c(6,7)]) - 0.3) #shift these left by -0.15
> boxplot(df[df$diet.type=="B", c(6,7)], xaxt = "n", add = TRUE, boxfill="lightblue",
+         boxwex=0.25, at = 1:ncol(df[,c(6,7)])  ) #shift to the right by +0.15
> boxplot(df[df$diet.type=="C", c(6,7)], xaxt = "n", add = TRUE, boxfill="orange",
+         boxwex=0.25, at = 1:ncol(df[,c(6,7)]) + 0.3) #shift these left by -0.15
> legend(x = "topleft",legend=c("Diet A", "Diet B", "Diet C"), fill = c("deeppink","lightblue","orange"))
```

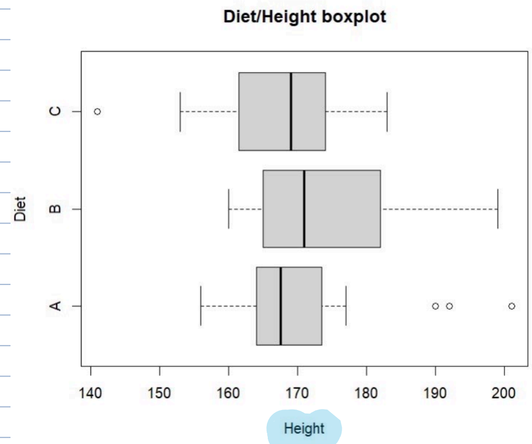
the weight loss in each group has a boxplot like this:




```
# Weight loss differnece in each group
df['difference']<-df$initial.weight-df$final.weight
boxplot(df$difference ~ df$diet.type ,main = "Weight loss in each group boxplot" ,xlab = "Weight loss", ylab="Diet",horizontal = TRUE)
```

the age and height boxplots are shown below:






```
boxplot(df$height ~ df$diet.type ,main = "Diet/Height boxplot" ,xlab = "Height", ylab="Diet",horizontal = TRUE)
boxplot(df$age ~ df$diet.type ,main = "Diet/Age boxplot" ,xlab = "Age", ylab="Diet",horizontal = TRUE)
```

b) Hypothesis:

H_0: The mean weight loss is the same across all the mentioned diet groups.

$$\mu_A = \mu_B = \mu_C$$

H_A: The mean weight loss differs between at least one pair of the mentioned diet groups.

Based on ANOVA test because p-value is small (less than α), reject H_0. meaning that, there is a significant difference in the mean weight loss between groups.

c) The ANOVA table is shown here:

```
> result <- aov(difference ~ diet.type, data = df)
> summary(result)
            Df Sum Sq Mean Sq F value Pr(>F)
diet.type    2   60.5  30.264   5.383 0.0066 **
Residuals   73  410.4   5.622
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

p-value < α. Therefore, with significance level of 0.05 we reject H_0. Therefore we can conclude that the mean weight loss differs between at least one pair of the mentioned diet groups.

d) Now in order to find which groups caused this significant difference we have to do a pairwise comparison with the below hypothesis for each pair of groups:

H_0: The mean weight loss in diet group i and diet group j is the same. $\mu_i - \mu_j = 0$

H_A: The mean weight loss in diet group i and diet group j is not the same.

$$\mu_i - \mu_j \neq 0$$

$$\alpha^* = \frac{\alpha}{3} = 0.0167$$

$$k = K(K-1)/2 = 3$$ تعداد مقایسات

```
> pairwise.t.test(df$difference, df$diet.type, p.adj = "none", pool.sd = FALSE)

        Pairwise comparisons using t tests with non-pooled SD

data:  df$difference and df$diet.type

  A      B
B 0.9622 -
C 0.0064 0.0076

P value adjustment method: none
```

• Group A and B:

H_0: The mean weight loss in diet group A and diet group B is the same: $\mu_A - \mu_B = 0$

H_A: The mean weight loss in diet group A and diet group B is not the same: $\mu_A - \mu_B \neq 0$

p-value in comparison between group A and B is 0.9622 which is bigger than $\alpha^* = 0.0167$ and we can not reject H_0.

• Group A and C:

H_0: The mean weight loss in diet group A and diet group C is the same: $\mu_A - \mu_C = 0$

H_A: The mean weight loss in diet group A and diet group C is not the same: $\mu_A - \mu_C \neq 0$

p-value in comparison between group A and C is 0.0064 which is smaller than $\alpha^* = 0.0167$ and we can reject H_0.

• Group B and C:

H_0: The mean weight loss in diet group B and diet group C is the same: $\mu_B - \mu_C = 0$

H_A: The mean weight loss in diet group B and diet group C is not the same: $\mu_B - \mu_C \neq 0$

p-value in comparison between group B and C is 0.0076 which is smaller than $\alpha^* = 0.0167$ and we can reject H_0.

(A-C and B-C)